# Density dependence

## Learning objectives

1. Understand how resource limitation and natural enemies affect population growth
2. Model density dependence with continuous time using a logistic model
3. Model density dependence with discrete time using a Beverton-Holt model and Ricker model

*review of last lecture: structured populations, but population growth rate wasn't affected by abundance in any stage*

*now we're adding density dependence, population growth depends on abundance*

## Density-dependent processes

*why would the number of individuals affect the growth of the population?*

### Resource limitation

- shared pool of resources *light, water, nutrients*
- as resource availability decreases, population growth decreases
- population has carrying capacity (K), equilibrium abundance

### Natural enemies

- herbivores, pathogens, seed predators
- as population size increases, attracts herbivores and predators, susceptible to disease
- survival decreases, back to equilibrium

### Self thinning

- as trees size increases, intercept light from seedlings, seedling mortality increases

*how do we model density dependence? again showing continuous and discrete time*

*start with continuous since more familiar: logistic equation*

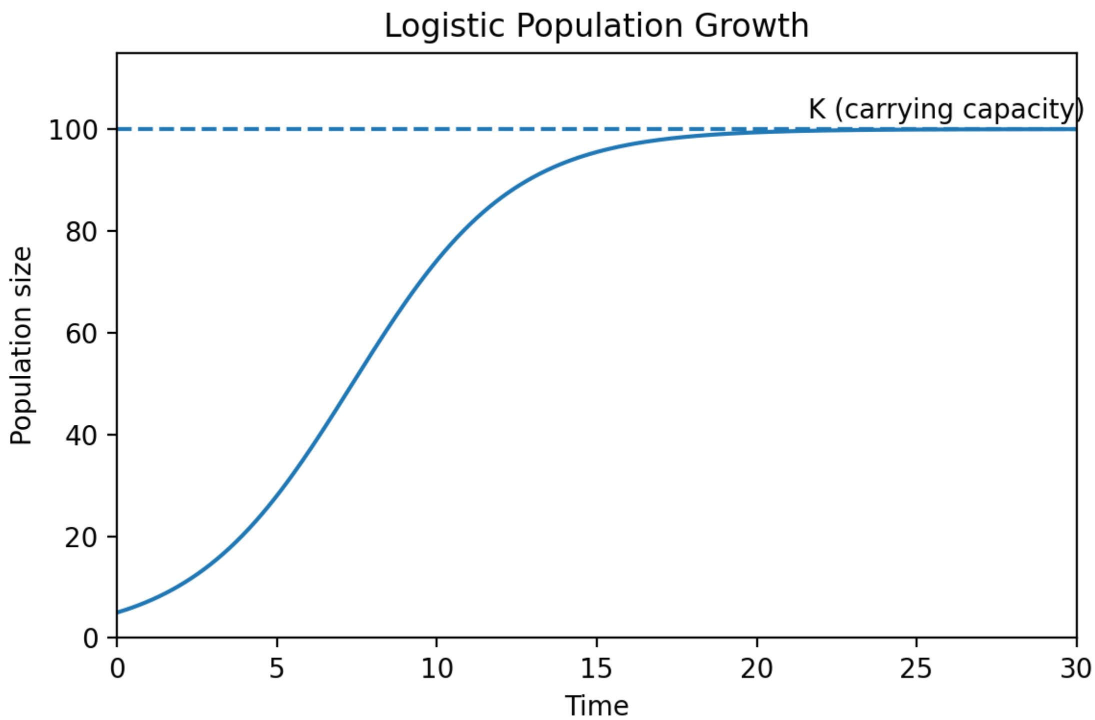
## Logistic equation

- r = intrinsic growth rate *same as r from first lecture*
- K = carrying capacity *equilibrium abundance*

$$\frac{dN}{dt} = rN(1 - \frac{N}{K})$$

*change in population size over time is rN, what we saw before*

*now limited by carrying capacity, so if $N = K$, no change in size*




*now let's go into some models for discrete time, which we'll play with in the lab and can be more common for analysis*

*two different models: Beverton-Holt and Ricker*

## Beverton-Holt model

- $\lambda$ - population growth rate
- $\alpha$ - competition *strength of competition between conspecifics*
- no K, related to $\alpha$

$$N_{t+1} = \frac{\lambda N_t}{1 + aN_t}$$

*again $\lambda N$ is that unrestricted population growth, but now it's limited by abundance with the competition scalar*

## Ricker model

- same variables

$$N_{t+1} = N_t \lambda e^{-\alpha N_t}$$

*same thing, $\lambda N$ but limited by abundance and competition coefficient*

**K vs. $\alpha$**

$$\alpha = \frac{\lambda - 1}{K}$$

*both discrete equations have basis in fisheries, where carrying capacity more helpful*

*in plant populations, more interested in that $\alpha$ for determing strength of density dependence and competition*